

# Cache

128 bytes

16 bytes / line

$128/16 = 8$  lines

00

→

01

→

10

→

11

→

16 bytes

2 bits idx

4 bits offset

4 bytes

32 bytes

4 bytes / line

2 lines per component

$32/4 = 8$  lines



tag

4 bits

idx

2 bits

offset

2 bits

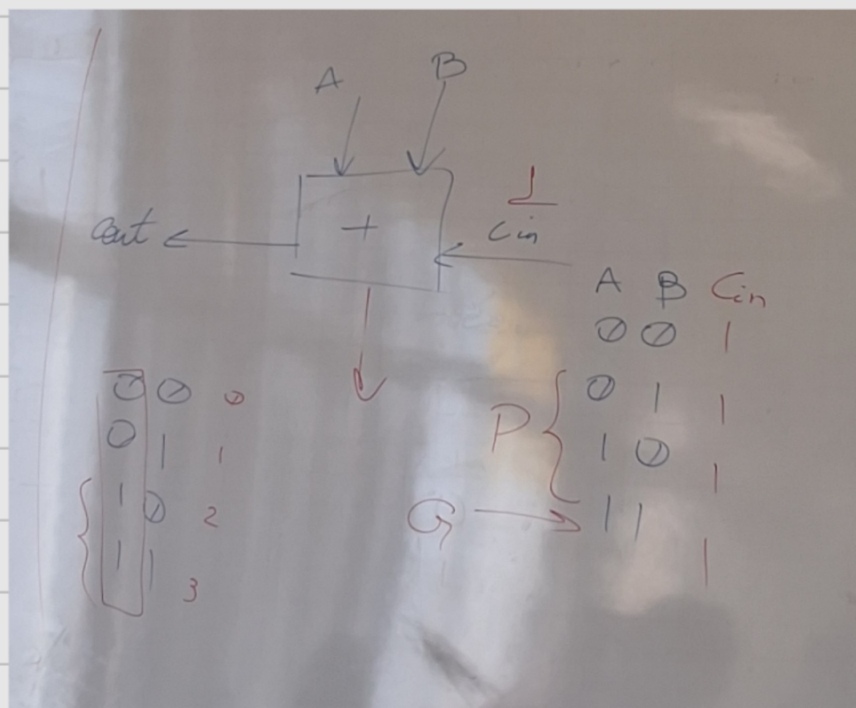
# álgebra booleana

$$\textcircled{7} f = (A+B)(A+B)$$

$$A(B+\bar{B})$$

$$A$$

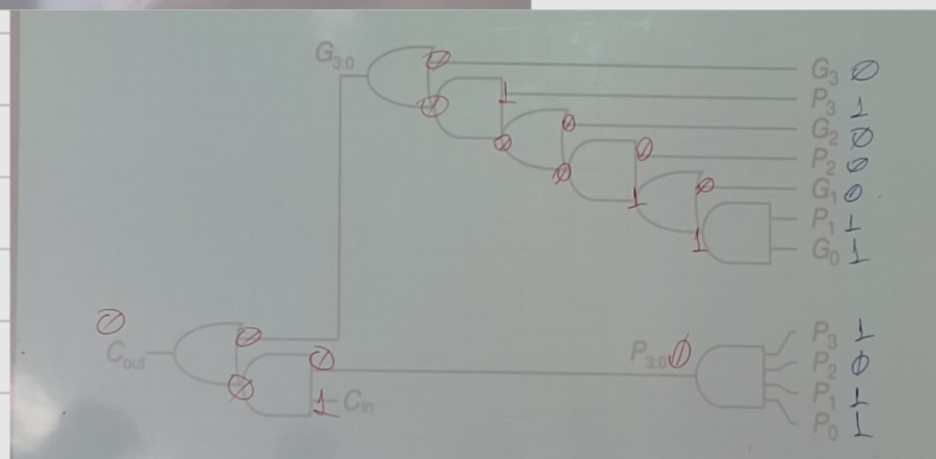
Ex 19



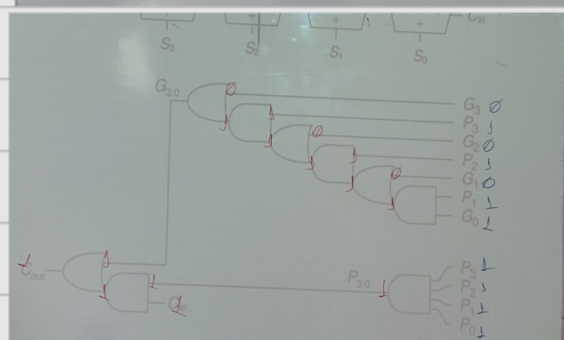
para gerar a soma  
tem que ser 1 na  
soma

1010

$$\begin{array}{r} 1011 \\ + 0001 \\ \hline \end{array} \quad C_{in}=1$$



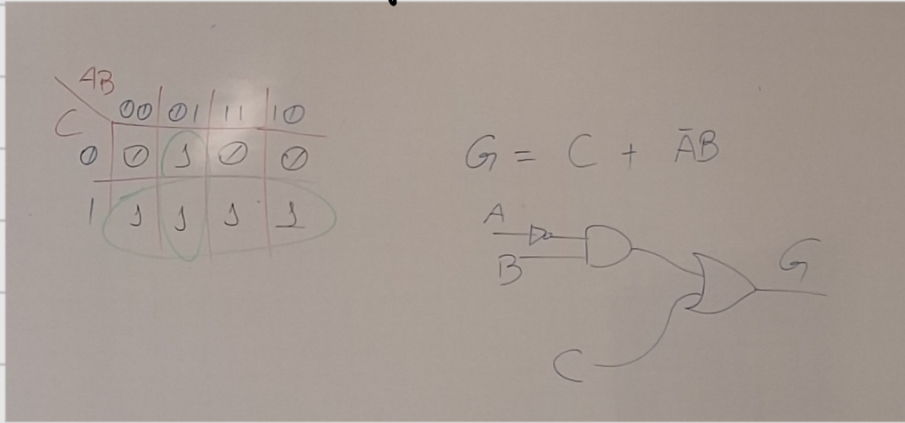
$$\begin{array}{r} 1011 \\ + 0101 \\ \hline \end{array} \quad C_{in}=1$$



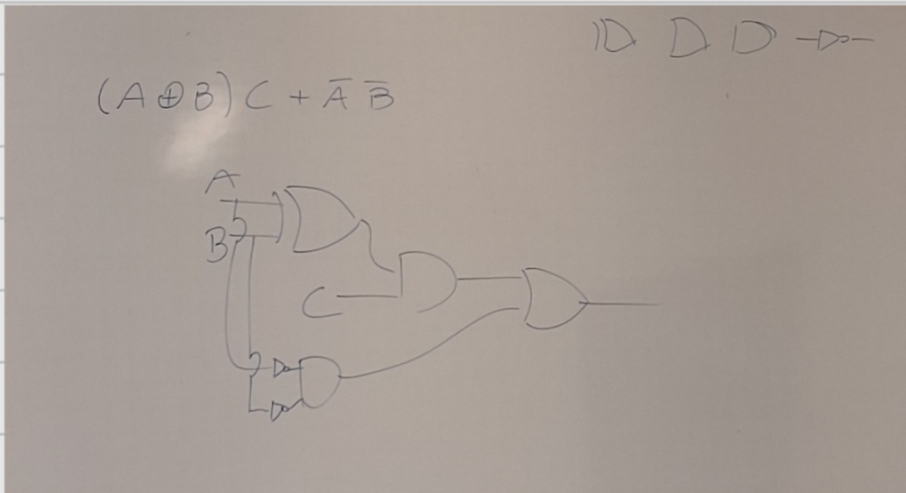
$$\begin{array}{r} 1011 \\ + 0101 \\ \hline \end{array} \quad C_{in}=0$$

$$\begin{array}{r} 0011 \\ 0111 \\ \hline \end{array} \quad C_{in}=1$$

Ex 1.2) Mappe de Karnaugh



Ex 1.4)



Ex 1.2) algèbre booléenne

$$G = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + \underline{A\bar{B}C + ABC}$$

$$\bar{A}(\bar{B}C + B\bar{C} + BC) + AC$$

$$\bar{A}(C + B\bar{C}) + AC$$

$$\bar{A}((C+B)(C+\bar{C})) + AC$$

$$\bar{A}C + \bar{A}B + AC$$

$$C + \bar{A}B$$

ou

$$G = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$

$$\bar{A}C + \bar{A}B\bar{C} + AC$$

$$C + \bar{A}B\bar{C}$$

$$(C + AB)(C + \bar{C})$$

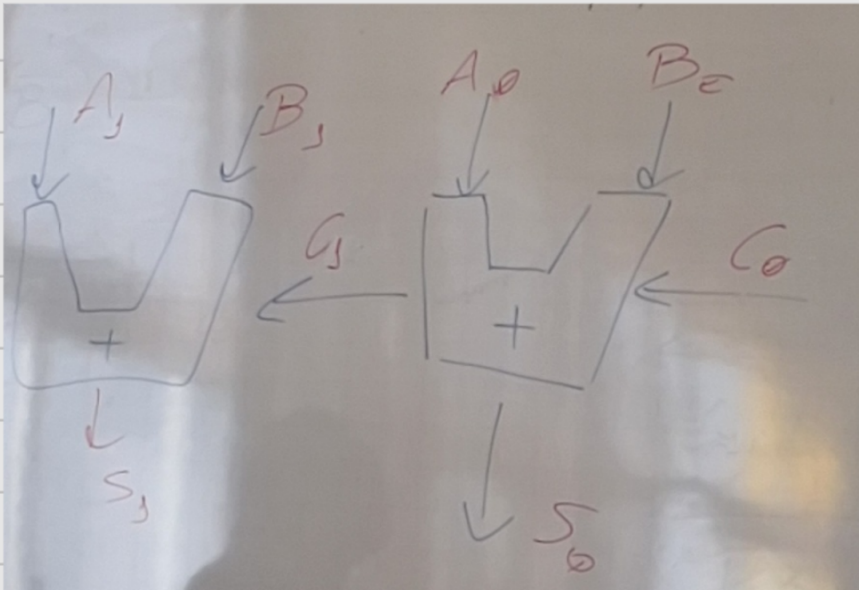
$$C + \bar{A}B$$

Ex (13)

| AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 | 1  | 0  | 0  | 1  |
| 01 | 1  | 0  | 0  | 1  |
| 11 | 1  | 0  | 1  | 1  |
| 10 | 1  | 0  | 1  | 1  |

$H = B + AC$

Ex (18)



Ex (23)

